Median of 2 sorted arrays -

arr1 → 7 12 14 15

arr2 → 1 2 3 4 9 11

Total length → 4 + 6 = 10, split = 10/2 = 5   Split → 5 and 5

Now,

low and high is based on the smaller length array, as log(small length) is much optimal

low → 0 (at min, 0 elements can be taken from 1st array)

high → arr length

Split 1 :-

l = 0 h = 4 m = 2

$$\begin{array}{ccc|ccc} & 7 & 12^{l_1} & r_1\,14 & 15 & \\ 1 & 2 & 3^{l_2} & r_2\,4 & 9 & 11 \end{array}$$

$l_2 < r_1$, $l_1 > r_2$ Not acceptable

Split 2 :-

l = 0 h = 1 m = 0

$$\begin{array}{ccccc|cccc} & & & & \_\,l_1 & r_1\,7 & 12 & 14 & 15 \\ 1 & 2 & 3 & 4 & 9^{l_2} & r_2\,11 & & & \end{array}$$

$l_1 < r_2$, $l_2 > r_1$ Not acceptable

Split 3 :-

l = 1 h = 1 m = 1

$$\begin{array}{cccc|ccc} & & & 7^{l_1} & r_1\,12 & 14 & 15 \\ 1 & 2 & 3 & 4^{l_2} & r_2\,9 & 11 & \end{array}$$

$\underbrace{l_1 < r_2,\ l_2 < r_1}_{\text{both less}}$ Acceptable

$$\boxed{\begin{array}{l} \text{splitAt} \to 5\ [\text{TotalLen}/2] \\ l_1 \to \text{arr1}[\text{mid}-1] \\ r_1 \to \text{arr2}[\text{mid}] \\ l_2 \to \text{arr1}[\text{splitAt}-\text{mid}-1] \\ r_2 \to \text{arr2}[\text{splitAt}-1] \end{array}}$$

Now,

$\max(l_1, l_2) \to 7$, $\min(r_1, r_2) \to 9$ → $\frac{7+9}{2} = \frac{16}{2} = 8$

arr1 → 7 12 14 15

arr2 → 1 2 3 4 9

Total length → 4 + 6 = 9, split = 9/2 = 4   Split → 4 and 5

Split 1 :-

l = 0 h = 4 m = 2

$$\begin{array}{cc|ccc} 7 & 12^{l_1} & r_1\,14 & 15 & \\ 1 & 2^{l_2} & r_2\,3 & 4 & 9 \end{array}$$

$l_2 < r_1$, $l_1 > r_2$ Not acceptable

Split 2 :-

l = 0 h = 1 m = 0

$$\begin{array}{cccc|cccc} & & & \_\,l_1 & r_1\,7 & 12 & 14 & 15 \\ 1 & 2 & 3 & 4^{l_2} & r_2\,9 & & & \end{array}$$

$\underbrace{l_1 < r_2,\ l_2 < r_1}_{\text{both less}}$ Acceptable

Split 3 :-

l = 1 h = 1 m = 1

$$\begin{array}{ccc|ccc} & & 7^{l_1} & r_1\,12 & 14 & 15 \\ 1 & 2 & 3^{l_2} & r_2\,4 & 9 & \end{array}$$

$l_1 < r_2$, $l_2 < r_1$ Not acceptable

Now, on the greater side, i.e on '5'th side,

$\min(r_2, r_1) \to \min(7, 9) \to 7$

$$\boxed{\begin{array}{l} TC \to O\log(\min(m,n)) \\ SC \to O(1) \end{array}}$$

$\boxed{\text{For finding element at 'K', splitAt will be 'K'}}$